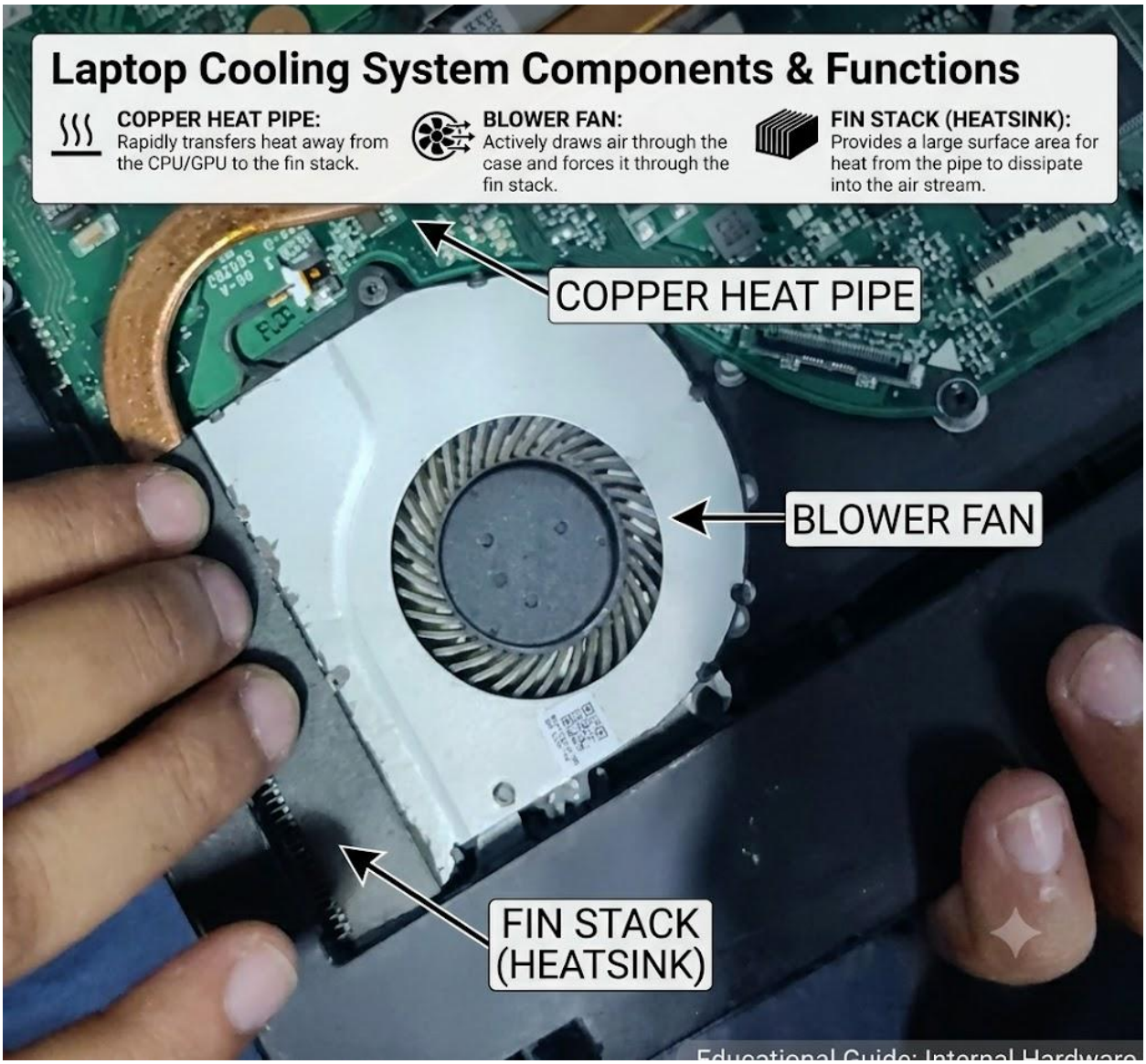
1) Open your own laptop or desktop (or use an online PC hardware simulator if you don't have access to a real system) and identify the GPU model installed. Write down the GPU's brand, model number, and whether it is integrated or dedicated.

Acer TravelMate P249 with an **Intel Core i3-6006U** processor

- **Brand:** Intel
- **Model Number:** Intel HD Graphics 520
- **Type:** Integrated (built directly into the processor die, sharing system RAM)

2) Take a clear photo or screenshot of the cooling system (fan, heatsink, or liquid cooling setup) inside your PC or from a PC hardware simulator. Label the main cooling components and briefly describe their function.

The Acer TravelMate P249 uses a standard air-cooling system with a single internal CPU cooling fan, as shown by multiple compatible 3-pin fan listings for the P249 models.



Copper Heat Pipe: Rapidly transfers thermal energy away from high-heat components like the CPU or GPU to the fin stack.

Blower Fan: Actively draws air through the system and forces a high-pressure stream of air through the fin stack.

Fin Stack (Heatsink): Provides an expanded surface area that absorbs heat from the copper pipe, allowing it to dissipate efficiently into the airflow.

3) Create a comparison table listing at least 3 differences between air cooling and liquid cooling systems for GPUs, including factors like cost, efficiency, and maintenance.

Here is a comparison table outlining the key differences between air cooling and liquid cooling systems for GPUs based on cost, efficiency, and maintenance:

Factor	Air Cooling System	Liquid Cooling System
Cost	Generally more affordable, making it a budget-friendly option for standard builds.	Significantly more expensive
Efficiency & Performance	Handles standard workloads and 1440p gaming effectively, though temperatures and fan noise can increase under sustained heavy loads.	Dissipates heat faster and maintains lower temperatures during long, intensive sessions, making it ideal for overclocking and 4K gaming.
Maintenance	Practically maintenance-free, requiring only occasional dusting of the heatsink and fans.	Requires routine monitoring
Acoustics (Noise)	Tends to produce more noise under heavy workloads as fans spin at higher speeds to dissipate heat.	Typically quieter because larger radiator fans can operate at lower RPMs while maintaining excellent cooling.

4) Imagine you are building a gaming PC for streaming IPL matches and playing graphics-heavy games. Choose between air cooling and liquid cooling for your GPU, and justify your choice in 3-4 sentences based on your research.

For a rig handling graphics-heavy games and live streaming, liquid cooling is the ideal choice for managing heavy workloads. It provides superior thermal efficiency and keeps high-end GPUs at lower temperatures during extended play sessions. Furthermore, it operates much more quietly under stress, ensuring fan noise won't interfere with your streaming setup.